Service Enumeration

It is essential to determine the application and its version as accurately as possible. We can use this information to scan for known vulnerabilities and analyze the source code for that version if we find it. An exact version number allows us to search for a more precise exploit that fits the service and the operating system of our target.

It is recommended to perform a quick port scan first, which gives us a small overview of the available ports. This causes significantly less traffic, which is advantageous for us because otherwise we can be discovered and blocked by the security mechanisms. We can deal with these first and run a port scan in the background, which shows all open ports (`-p-`).We can use the version scan to scan the specific ports for services and their versions (`-sV`).

A full port scan takes quite a long time. To view the scan status, we can press the `[Space Bar]` during the scan, which will cause `Nmap` to show us the scan status.

```
pecna@htb[/htb]$ sudo nmap 10.129.2.28 -p- -sV

Starting Nmap 7.80 ( https://nmap.org ) at 2020-06-15 19:44
CEST
[Space Bar]
Stats: 0:00:03 elapsed; 0 hosts completed (1 up), 1 undergo
ing SYN Stealth Scan
SYN Stealth Scan Timing: About 3.64% done; ETC: 19:45 (0:0
0:53 remaining)
```

Scanning Options	Description
`10.129.2.28`	Scans the specified target.
`-p-`	Scans all ports.
`-sV`	Performs service version detection on specified ports.

Another option (`--stats-every=5s`) that we can use is defining how periods of time the status should be shown. Here we can specify the number of seconds (`s`) or minutes (`m`), after which we want to get the status.

```
pecna@htb[/htb]$ sudo nmap 10.129.2.28 -p- -sV --stats-ever
y=5sStarting Nmap 7.80 ( https://nmap.org ) at 2020-06-15 1
9:46 CEST
Stats: 0:00:05 elapsed; 0 hosts completed (1 up), 1 undergo
ing SYN Stealth Scan
SYN Stealth Scan Timing: About 13.91% done; ETC: 19:49 (0:0
0:31 remaining)
Stats: 0:00:10 elapsed; 0 hosts completed (1 up), 1 undergo
ing SYN Stealth Scan
SYN Stealth Scan Timing: About 39.57% done; ETC: 19:48 (0:0
0:15 remaining)
```

Scanning Options	Description
`10.129.2.28`	Scans the specified target.
`-p-`	Scans all ports.
`-sV`	Performs service version detection on specified ports.
`--stats-every=5s`	Shows the progress of the scan every 5 seconds.

We can also increase the `verbosity level` (`-v` / `-vv`), which will show us the open ports directly when `Nmap` detects them.

Service Enumeration

```
pecna@htb[/htb]$ sudo nmap 10.129.2.28 -p- -sV -v Starting
Nmap 7.80 ( https://nmap.org ) at 2020-06-15 20:03 CEST
NSE: Loaded 45 scripts for scanning.
Initiating ARP Ping Scan at 20:03
Scanning 10.129.2.28 [1 port]
Completed ARP Ping Scan at 20:03, 0.03s elapsed (1 total ho
sts)
Initiating Parallel DNS resolution of 1 host. at 20:03
Completed Parallel DNS resolution of 1 host. at 20:03, 0.02
s elapsed
Initiating SYN Stealth Scan at 20:03
Scanning 10.129.2.28 [65535 ports]
Discovered open port 995/tcp on 10.129.2.28
Discovered open port 80/tcp on 10.129.2.28
Discovered open port 993/tcp on 10.129.2.28
```

```
Discovered open port 143/tcp on 10.129.2.28
Discovered open port 25/tcp on 10.129.2.28
Discovered open port 110/tcp on 10.129.2.28
Discovered open port 22/tcp on 10.129.2.28
<SNIP>
```

Scanning Options	Description
`10.129.2.28`	Scans the specified target.
`-p-`	Scans all ports.
`-sV`	Performs service version detection on specified ports.
`-v`	Increases the verbosity of the scan, which displays more detailed information.

Once the scan is complete, we will see all TCP ports with the corresponding service and their versions that are active on the system.

```
pecna@htb[/htb]$ sudo nmap 10.129.2.28 -p- -sV

Starting Nmap 7.80 ( https://nmap.org ) at 2020-06-15 20:00
CEST
Nmap scan report for 10.129.2.28
Host is up (0.013s latency).
Not shown: 65525 closed ports
PORT      STATE    SERVICE      VERSION
22/tcp    open     ssh          OpenSSH 7.6p1 Ubuntu 4ubunt
u0.3 (Ubuntu Linux; protocol 2.0)
25/tcp    open     smtp         Postfix smtpd
80/tcp    open     http         Apache httpd 2.4.29 ((Ubunt
u))
110/tcp   open     pop3         Dovecot pop3d
139/tcp   filtered netbios-ssn
143/tcp   open     imap         Dovecot imapd (Ubuntu)
445/tcp   filtered microsoft-ds
993/tcp   open     ssl/imap     Dovecot imapd (Ubuntu)
995/tcp   open     ssl/pop3     Dovecot pop3d
MAC Address: DE:AD:00:00:BE:EF (Intel Corporate)
Service Info: Host:  inlane; OS: Linux; CPE: cpe:/o:linux:l
```

```
inux_kernel

Service detection performed. Please report any incorrect re
sults at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 91.73 second
s
```

Scanning Options	Description
`10.129.2.28`	Scans the specified target.
`-p-`	Scans all ports.
`-sV`	Performs service version detection on specified ports.

Primarily, `Nmap` looks at the banners of the scanned ports and prints them out. If it cannot identify versions through the banners, `Nmap` attempts to identify them through a signature-based matching system, but this significantly increases the scan's duration. One disadvantage to `Nmap`'s presented results is that the automatic scan can miss some information because sometimes `Nmap` does not know how to handle it. Let us look at an example of this.

```
pecna@htb[/htb]$ sudo nmap 10.129.2.28 -p- -sV -Pn -n --dis
able-arp-ping --packet-traceStarting Nmap 7.80 ( https://nm
ap.org ) at 2020-06-16 20:10 CEST
<SNIP>
NSOCK INFO [0.4200s] nsock_trace_handler_callback(): Callba
ck: READ SUCCESS for EID 18 [10.129.2.28:25] (35 bytes): 22
0 inlane ESMTP Postfix (Ubuntu)..
Service scan match (Probe NULL matched with NULL line 310
4): 10.129.2.28:25 is smtp.  Version: |Postfix smtpd|||
NSOCK INFO [0.4200s] nsock_iod_delete(): nsock_iod_delete
(IOD#1)Nmap scan report for 10.129.2.28
Host is up (0.076s latency).

PORT   STATE SERVICE VERSION
25/tcp open  smtp    Postfix smtpd
MAC Address: DE:AD:00:00:BE:EF (Intel Corporate)
Service Info: Host:  inlane

Service detection performed. Please report any incorrect re
```

```
sults at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.47 seconds
```

Scanning Options	Description
`10.129.2.28`	Scans the specified target.
`-p-`	Scans all ports.
`-sV`	Performs service version detection on specified ports.
`-Pn`	Disables ICMP Echo requests.
`-n`	Disables DNS resolution.
`--disable-arp-ping`	Disables ARP ping.
`--packet-trace`	Shows all packets sent and received.

If we look at the results from `Nmap`, we can see the port's status, service name, and hostname. Nevertheless, let us look at this line here:

- `NSOCK INFO [0.4200s] nsock_trace_handler_callback(): Callback: READ SUCCESS for EID 18 [10.129.2.28:25] (35 bytes): 220 inlane ESMTP Postfix (Ubuntu)..`

Then we see that the SMTP server on our target gave us more information than `Nmap` showed us. Because here, we see that it is the Linux distribution `Ubuntu`. It happens because, after a successful three-way handshake, the server often sends a banner for identification. This serves to let the client know which service it is working with. At the network level, this happens with a `PSH` flag in the TCP header. However, it can happen that some services do not immediately provide such information. It is also possible to remove or manipulate the banners from the respective services. If we `manually` connect to the SMTP server using `nc`, grab the banner, and intercept the network traffic using `tcpdump`, we can see what `Nmap` did not show us.

Tcpdump

```
pecna@htb[/htb]$ sudo tcpdump -i eth0 host 10.10.14.2 and 1
0.129.2.28

tcpdump: verbose output suppressed, use -v or -vv for full
protocol decode
```

```
listening on eth0, link-type EN10MB (Ethernet), capture siz
e 262144 bytes
```

Nc

```
pecna@htb[/htb]$  nc -nv 10.129.2.28 25

Connection to 10.129.2.28 port 25 [tcp/*] succeeded!
220 inlane ESMTP Postfix (Ubuntu)
```

Tcpdump - Intercepted Traffic

```
18:28:07.128564 IP 10.10.14.2.59618 > 10.129.2.28.smtp: Fla
gs [S], seq 1798872233, win 65535, options [mss 1460,nop,ws
cale 6,nop,nop,TS val 331260178 ecr 0,sackOK,eol], length 0

18:28:07.255151 IP 10.129.2.28.smtp > 10.10.14.2.59618: Fla
gs [S.], seq 1130574379, ack 1798872234, win 65160, options
[mss 1460,sackOK,TS val 1800383922 ecr 331260178,nop,wscale
7], length 0

18:28:07.255281 IP 10.10.14.2.59618 > 10.129.2.28.smtp: Fla
gs [.], ack 1, win 2058, options [nop,nop,TS val 331260304
ecr 1800383922], length 0

18:28:07.319306 IP 10.129.2.28.smtp > 10.10.14.2.59618: Fla
gs [P.], seq 1:36, ack 1, win 510, options [nop,nop,TS val
1800383985 ecr 331260304], length 35: SMTP: 220 inlane ESMT
P Postfix (Ubuntu)

18:28:07.319426 IP 10.10.14.2.59618 > 10.129.2.28.smtp: Fla
gs [.], ack 36, win 2058, options [nop,nop,TS val 331260368
ecr 1800383985], length 0
```

The first three lines show us the three-way handshake.

1.	`[SYN]`	`18:28:07.128564 IP 10.10.14.2.59618 > 10.129.2.28.smtp: Flags [S], <SNIP>`

2.	`[SYN-ACK]`	`18:28:07.255151 IP 10.129.2.28.smtp > 10.10.14.2.59618: Flags [S.], <SNIP>`
3.	`[ACK]`	`18:28:07.255281 IP 10.10.14.2.59618 > 10.129.2.28.smtp: Flags [.], <SNIP>`

After that, the target SMTP server sends us a TCP packet with the `PSH` and `ACK` flags, where `PSH` states that the target server is sending data to us and with `ACK` simultaneously informs us that all required data has been sent.

4. `[PSH-ACK]`	`18:28:07.319306 IP 10.129.2.28.smtp > 10.10.14.2.59618: Flags [P.], <SNIP>`	

The last TCP packet that we sent confirms the receipt of the data with an `ACK`

5.	`[ACK]`	`18:28:07.319426 IP 10.10.14.2.59618 > 10.129.2.28.smtp: Flags [.], <SNIP>`